Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

DEPARTMENT OF MECHATRONICS

VII SEMESTER B.TECH. (MECHATRONICS)

END SEMESTER EXAMINATIONS, NOV-2022

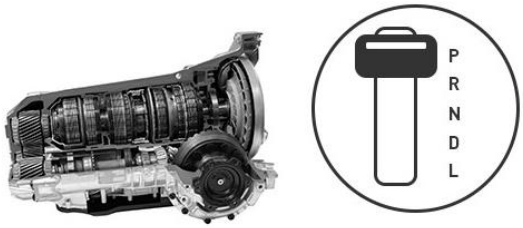
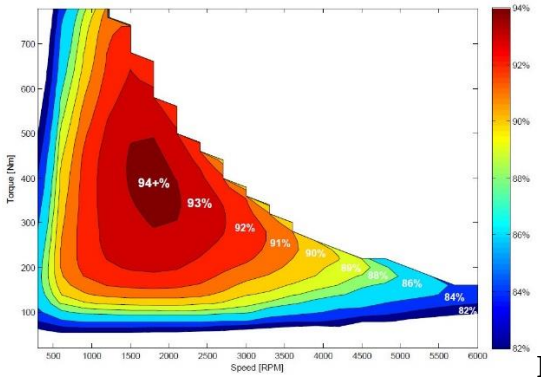
SUBJECT: MODELING OF ELECTRIC VEHICLES [MTE 4085]

(30-11-2022)

Time: 3 Hours **MAX. MARKS: 50**

Instructions to Candidates:

- Answer **ALL** the questions.

Q. No		M	CO	PO	LO	BL
1A.	Identify the type of transmission shown in Fig 1A and outline the key features in contrast to the counterpart.  **Fig 1A.**	**3**	**1**	**1**	**1**	**3**
1B.	Examine modeling approaches with suitable examples.	**3**	**1**	**5**	**3**	**4**
1C.	Make use of block diagram to depict ICE vehicle architectures.	**4**	**1**	**1**	**1**	**3**
2A.	Inspect the role of design consideration for aerodynamic drag and transmission in Electric Vehicles.	**4**	**2**	**3**	**2**	**4**
2B.	Analyze the safety significance of 3Box Model of vehicle	**3**	**2**	**6**	**2**	**4**
2C.	List the specific requirements of chassis/Body.	**3**	**2**	**2**	**1**	**4**
3A.	Model the Integro-differential equation of DC excited RL circuit.	**3**	**2**	**5**	**3**	**3**
3B.	Identify the type of motor technology from the efficiency map as shown in Fig 3B with justification.  **Fig 3B.**	**2**	**2**	**1**	**1**	**3**

3C.	Model the PMSM Motor with relevant equivalent governing equations.	**5**	**3**	**5**	**3**	**3**
4A.	Compare the key advantages and disadvantages of reluctance based motors over PMSM.	**3**	**3**	**1**	**1**	**4**
4B.	Examine the control of a 3HP, 1500RPM, separately excited DC Motor for EV. Motor rated voltage is 230V and field voltage of 300V.	**3**	**3**	**1**	**1**	**3**
4C.	Inspect the vector control of Induction motor.	**4**	**3**	**1**	**1**	**4**
5A.	List the key roles of vehicle dynamic control as fourth element.	**3**	**3**	**1**	**1**	**4**
5B.	Compare the active and passive cell balancing techniques.	**3**	**5**	**1**	**1**	**4**
5C.	Model the Battery electrical and thermal systems.	**4**	**5**	**5**	**3**	**3**